

NUMERISCHE MATHEMATIK I | HAUPTKOMONENTENANALYSE

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INTRODUCTION

In this exercise we consider the MNIST data set. It consists of 60,000 grayscale images of size 28×28 . Each image shows a handwritten digit from 0 to 9. In addition, each image has a label indicating which digit is shown in the image. In the following, our goal is to classify images using machine learning methods.



The MNIST data set is divided into 50,000 *training* samples and 10,000 *test* samples. The idea is to use the training data (images and labels) to construct our algorithm. Afterwards, we use the test data to evaluate how well the method performs.

The algorithm considered in this programming exercise consists of two steps. First, we **reduce the dimension using Principal Component Analysis (PCA)**. Then we **perform classification using the K-means algorithm**.

Input Remark. Remark. The images of the training data set can be read in Python as follows:

```
imgs = np.fromfile('train-images.idx3-ubyte', dtype=np.uint8)
imgs = imgs.reshape((16000, [-1, 28, 28]))
```

```
labs = np.fromfile('train-labels.idx3-ubyte', dtype=np.uint8)
labs = labs[8:]
```

1. PRINCIPAL COMPONENT ANALYSIS

We begin with dimensionality reduction. Let N be large and let $x_0, \dots, x_{N-1} \in \mathbb{R}^D$ be given points with large ambient dimension D . The goal is to find an affine subspace of small dimension $d \ll D$ such that the sum of squared distances of the points to this subspace is minimal.

Definition 1.1. An affine subspace of $\mathbb{R}^D$ of dimension d is a set

$$H_{A,b} := \{At + b : t \in \mathbb{R}^d\},$$

where $A \in \mathbb{R}^{D \times d}$ has orthonormal columns (i.e. $A^T A = I_d$) and $b \in \mathbb{R}^D$ is arbitrary.

For a point $x \in \mathbb{R}^D$ we define the squared distance to a set M by

$$d^2(x, M) := \inf_{y \in M} \|x - y\|_2^2.$$

Lemma 1.2. *Let $x \in \mathbb{R}^D$ and let $H_{A,b}$ be an affine subspace. Then*

$$d^2(x, H_{A,b}) = \|\text{proj}_{H_{A,b}}(x) - x\|_2^2,$$

where

$$\text{proj}_{H_{A,b}}(x) = AA^T(x - b) + b$$

denotes the orthogonal projection of x onto $H_{A,b}$.

Beweis. Let $y = At + b \in H_{A,b}$. Minimizing the quadratic function in t (by setting the first derivative equal to zero) yields $A^T(At + b - x) = 0$, hence $A^TAt = A^T(x - b)$. Since $A^TA = I_d$, we obtain $t = A^T(x - b)$. Substituting this expression gives $\text{proj}_{H_{A,b}}(x) = AA^T(x - b) + b$. $\square$

Problem 1.3. Given $x_0, \dots, x_{N-1} \in \mathbb{R}^D$, find $A \in \mathbb{R}^{D \times d}$ and $b \in \mathbb{R}^D$ with $A^TA = I_d$ such that

$$(\hat{A}, \hat{b}) \in \underset{A, b}{\operatorname{argmin}} \sum_{i=0}^{N-1} \|\text{proj}_{H_{A,b}}(x_i) - x_i\|_2^2.$$

Proposition 1.4. *An optimal pair $(\hat{A}, \hat{b})$ is given as follows. The mean*

$$\hat{b} = \frac{1}{N} \sum_{i=0}^{N-1} x_i$$

is optimal, and the columns of $\hat{A} = (\hat{a}_0, \dots, \hat{a}_{d-1})$ are the eigenvectors corresponding to the d largest eigenvalues of

$$S = YY^T = \sum_{i=0}^{N-1} (x_i - \hat{b})(x_i - \hat{b})^T,$$

the empirical covariance matrix. (In short: $\hat{A}$ contains the directions of largest variance.)

Beweis. After suitable reformulations (see standard PCA results), the objective function can be written as

$$\sum_{i=0}^{N-1} \|(I - AA^T)(x_i - b)\|_2^2.$$

Differentiating with respect to b and setting the derivative equal to zero shows that the mean $\hat{b}$ satisfies the optimality condition.

For the minimization with respect to A for fixed $b = \hat{b}$, define $y_i = x_i - \hat{b}$ and $Y = (y_0 | \dots | y_{N-1})$. Then

$$\sum_{i=0}^{N-1} \|(I - AA^T)y_i\|_2^2 = \sum_{i=0}^{N-1} \left(\|y_i\|_2^2 - \sum_{j=0}^{d-1} (a_j^T y_i)^2 \right).$$

Minimizing this expression is equivalent to maximizing the sum of the projection terms, which yields the d eigenvectors of $S = YY^T$ corresponding to the largest eigenvalues. $\square$

Lemma 1.5. *Let $Y \in \mathbb{R}^{D \times N}$ have the singular value decomposition $Y = U\Sigma V^T$. Then the matrix $S = YY^T$ has the eigenvalue decomposition*

$$S = U\Sigma^2U^T,$$

i.e. the left singular vectors of Y are the eigenvectors of S .

2. THE K -MEANS ALGORITHM

After dimensionality reduction we work in $\mathbb{R}^d$ with the low-dimensional representations. The K -means algorithm partitions N points $x_0, \dots, x_{N-1} \in \mathbb{R}^d$ into K clusters, represented by centers $r_0, \dots, r_{K-1} \in \mathbb{R}^d$.

Bemerkung 2.1 (Idea). Each point is assigned to the nearest center, and each center is the mean of the points assigned to it. A standard iteration consists of:

- (1) assigning points to clusters,
- (2) updating the cluster centers.

Algorithm 1 K-Means

Require: Points $x_0, \dots, x_{N-1} \in \mathbb{R}^d$, number of clusters K

```

1: Initialize centers  $r_0, \dots, r_{K-1}$  (e.g. randomly)
2: while stopping criterion not satisfied do
3:   for  $i = 0$  to  $N - 1$  do
4:      $k^* \leftarrow \arg \min_k \|x_i - r_k\|_2^2$ 
5:     Assign  $i$  to cluster  $C_{k^*}$ 
6:   end for
7:   for  $k = 0$  to  $K - 1$  do
8:      $r_k \leftarrow \frac{1}{|C_k|} \sum_{i \in C_k} x_i$ 
9:   end for
10: end while
    
```

Bemerkung 2.2. For the K -means algorithm we do not require the labels of the training images. This is called *unsupervised learning*. Algorithms that use labels are called *supervised learning*.

3. ECKART–YOUNG–MIRSKY THEOREM

Theorem 3.1 (Eckart–Young–Mirsky). *Let $\mathbf{A} \in \mathbb{K}^{M \times N}$ be a matrix of rank r with singular value decomposition*

$$\mathbf{A} = \mathbf{U} \Sigma \mathbf{V}^*,$$

where the singular values satisfy $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$. Then for $k \leq r$ the best rank- k approximation is given by

$$\mathbf{A}_k \in \arg \min_{\text{rank}(\mathbf{B})=k} \|\mathbf{A} - \mathbf{B}\|_2 = \arg \min_{\text{rank}(\mathbf{B})=k} \|\mathbf{A} - \mathbf{B}\|_F,$$

with

$$\mathbf{A}_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^*.$$

Moreover,

$$\min_{\text{rank}(\mathbf{B})=k} \|\mathbf{A} - \mathbf{B}\|_2 = \sigma_{k+1}, \quad \min_{\text{rank}(\mathbf{B})=k} \|\mathbf{A} - \mathbf{B}\|_F = \left(\sum_{j=k+1}^r \sigma_j^2 \right)^{1/2}.$$

Beweis. We prove the claim for the spectral norm. The argument for the Frobenius norm follows analogously, using that both the spectral norm and the Frobenius norm are invariant under multiplication by orthogonal/unitary matrices.

Clearly,

$$\text{rank}(\mathbf{A}_k) = k.$$

We compute

$$\begin{aligned}\|\mathbf{A} - \mathbf{A}_k\|_2 &= \|\mathbf{U}^*(\mathbf{A} - \mathbf{A}_k)\mathbf{V}\|_2 \\ &= \|\text{diag}(0, \dots, 0, \sigma_{k+1}, \dots, \sigma_r)\|_2 = \sigma_{k+1}.\end{aligned}$$

We now show that any other matrix $\mathbf{B} \in \mathbb{R}^{M \times N}$ with $\text{rank}(\mathbf{B}) = k$ yields a worse approximation of $\mathbf{A}$ in the spectral norm. Since $\text{rank}(\mathbf{B}) = k$, there exist orthonormal vectors $\mathbf{x}_1, \dots, \mathbf{x}_{N-k}$ such that

$$\mathcal{N}(\mathbf{B}) = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_{N-k}\}.$$

By a dimension argument ($N - k + (k + 1) = N + 1$), we have

$$\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_{N-k}\} \cap \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{k+1}\} \neq \{\mathbf{0}\}.$$

Let

$$\mathbf{z} \in \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_{N-k}\} \cap \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{k+1}\}, \quad \|\mathbf{z}\|_2 = 1.$$

Then $\mathbf{B}\mathbf{z} = \mathbf{0}$. Moreover,

$$\mathbf{z} = \sum_{j=1}^{k+1} (\mathbf{v}_j^T \mathbf{z}) \mathbf{v}_j, \quad \sum_{j=1}^{k+1} (\mathbf{v}_j^T \mathbf{z})^2 = 1.$$

Consequently,

$$\mathbf{A}\mathbf{z} = \sum_{j=1}^{k+1} \sigma_j (\mathbf{v}_j^T \mathbf{z}) \mathbf{u}_j.$$

Since the spectral norm is compatible with the Euclidean vector norm, we obtain

$$\begin{aligned}\|\mathbf{A} - \mathbf{B}\|_2^2 &\geq \|(\mathbf{A} - \mathbf{B})\mathbf{z}\|_2^2 = \|\mathbf{A}\mathbf{z}\|_2^2 \\ &= \sum_{j=1}^{k+1} \sigma_j^2 (\mathbf{v}_j^T \mathbf{z})^2 \geq \sigma_{k+1}^2.\end{aligned}$$

This proves optimality with respect to the spectral norm.

It remains to establish the result for the Frobenius norm. For this we use the following singular value estimate.

Remark. For arbitrary matrices $\mathbf{A}, \mathbf{B} \in \mathbb{K}^{M \times N}$,

$$\sigma_{i+j-1}(\mathbf{A} + \mathbf{B}) \leq \sigma_i(\mathbf{A}) + \sigma_j(\mathbf{B}),$$

where $\sigma_\ell \equiv 0$ for $\ell > \min\{M, N\}$.

Proof of the Remark. Let $\mathbf{C} = \mathbf{A} + \mathbf{B}$ and let $\mathbf{C}_{i+j-2}$, $\mathbf{A}_{i-1}$, and $\mathbf{B}_{j-1}$ denote best rank- $(i+j-2)$, rank- $(i-1)$, and rank- $(j-1)$ approximations of $\mathbf{C}$, $\mathbf{A}$, and $\mathbf{B}$, respectively. Then

$$\begin{aligned}\sigma_{i+j-1}(\mathbf{C}) &= \sigma_1(\mathbf{C} - \mathbf{C}_{i+j-2}) \\ &\leq \sigma_1(\mathbf{C} - \mathbf{A}_{i-1} - \mathbf{B}_{j-1}) \\ &= \sigma_1(\mathbf{A} - \mathbf{A}_{i-1} + \mathbf{B} - \mathbf{B}_{j-1}) \\ &\leq \sigma_1(\mathbf{A} - \mathbf{A}_{i-1}) + \sigma_1(\mathbf{B} - \mathbf{B}_{j-1}) \\ &= \sigma_i(\mathbf{A}) + \sigma_j(\mathbf{B}).\end{aligned}$$

The first inequality follows since $\text{rank}(\mathbf{A}_{i-1} + \mathbf{B}_{j-1}) \leq i+j-2$, and the second from the triangle inequality. $\circ$

Now let $\text{rank}(\mathbf{B}) \leq k$. Then $\sigma_j(\mathbf{B}) = 0$ for all $j \geq k+1$, and by the remark,

$$\sigma_{i+k}(\mathbf{A}) \leq \sigma_i(\mathbf{A} - \mathbf{B}).$$

Hence,

$$\begin{aligned}\|\mathbf{A} - \mathbf{B}\|_F^2 &= \sum_{j=1}^r \sigma_j(\mathbf{A} - \mathbf{B})^2 \\ &\geq \sum_{j=1}^r \sigma_{j+k}(\mathbf{A})^2 = \sum_{j=k+1}^r \sigma_j(\mathbf{A})^2 = \|\mathbf{A} - \mathbf{A}_k\|_F^2.\end{aligned}$$

This shows that $\mathbf{A}_k$ is also the optimal rank- k approximation with respect to the Frobenius norm, and that the inequality is strict whenever $\mathbf{B} \neq \mathbf{A}_k$. $\square$

We now consider a related statement concerning approximation by an orthogonal matrix, i.e. for real square matrices.

Theorem 3.2. *Let $\mathbf{A} \in \mathbb{K}^{N \times N}$ be a real square matrix with singular value decomposition $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T$. Then*

$$\mathbf{U}\mathbf{V}^T \in \arg \min_{\mathbf{R} \in O(N)} \|\mathbf{A} - \mathbf{R}\|_F.$$

Beweis. We compute

$$\|\mathbf{A} - \mathbf{R}\|_F = \|\mathbf{U}^T(\mathbf{A} - \mathbf{R})\mathbf{V}\|_F = \|\mathbf{D} - \mathbf{U}^T\mathbf{R}\mathbf{V}\|_F,$$

where $\mathbf{U}^T\mathbf{R}\mathbf{V} \in O(N)$. Thus, the problem reduces to approximating a diagonal matrix by an orthogonal one, with the optimal choice being $\mathbf{I}_N$.

Indeed,

$$\begin{aligned}\|\mathbf{D} - \mathbf{R}\|_F^2 &= \text{tr}((\mathbf{D} - \mathbf{R})^T(\mathbf{D} - \mathbf{R})) \\ &= \text{tr}(\mathbf{D}^2) - 2\text{tr}(\mathbf{D}\mathbf{R}) + \text{tr}(\mathbf{R}^T\mathbf{R}) \\ &= \|\mathbf{D}\|_F^2 - 2\text{tr}(\mathbf{D}\mathbf{R}) + N.\end{aligned}$$

Hence, minimization is equivalent to maximizing $\text{tr}(\mathbf{D}\mathbf{R})$. Since the entries of an orthogonal matrix satisfy $|r_{ij}| \leq 1$, we have

$$\text{tr}(\mathbf{D}\mathbf{R}) = \sum_{j=1}^N r_{jj}\sigma_j \leq \sum_{j=1}^N \sigma_j = \text{tr}(\mathbf{D}).$$

Therefore,

$$\|\mathbf{D} - \mathbf{R}\|_F^2 \geq \|\mathbf{D}\|_F^2 - 2\text{tr}(\mathbf{D}) + N = \|\mathbf{D} - \mathbf{I}_N\|_F^2,$$

which proves the claim. $\square$